

WIA2007 Mobile Application Development

Final Project Report

Project Title

ClimateBuddy: A Climate Action Awareness and Carbon Footprint Tracking Mobile Application

1. Introduction and Problem Context

Climate change is one of the most critical global challenges affecting environmental sustainability, public health, and future generations. Many individuals are unaware of how their daily activities contribute to carbon emissions and environmental degradation. This lack of awareness reduces personal accountability and slows down collective climate action.

This project supports **Sustainable Development Goal (SDG) 13: Climate Action**, specifically **Target 13.3**, which focuses on improving education, awareness, and capacity on climate change mitigation and impact reduction.

ClimateBuddy is a mobile application designed to raise awareness about climate change and help users understand, track, and reduce their personal carbon footprint through simple daily actions, educational content, and progress tracking.

2. System Requirements

2.1 Findings from User Study

A simple user study was conducted through informal interviews and online discussions with students and young adults. The findings revealed that: - Many users are aware of climate change but do not know how their daily activities contribute to it. - Users prefer simple interfaces and clear guidance rather than complex scientific data. - Motivation features such as progress tracking and tips are helpful to encourage sustainable behavior.

These findings guided the design of ClimateBuddy to be simple, informative, and user-friendly.

2.2 Functional Requirements

The system consists of **five main modules**, each with three functional requirements.

Module 1: User Authentication

1. The system shall allow users to register using email and password.
2. The system shall allow users to log in securely.
3. The system shall allow users to log out from the application.

Module 2: Carbon Footprint Calculator

1. The system shall allow users to input daily activities such as transportation and energy usage.
2. The system shall calculate estimated carbon emissions based on user input.
3. The system shall display the calculated carbon footprint in a clear format.

Module 3: Daily Tracking and History

1. The system shall store daily carbon footprint records.
2. The system shall allow users to view past records.
3. The system shall display trends using simple visual indicators.

Module 4: Climate Education and Tips

1. The system shall display educational content related to climate change.
2. The system shall provide daily tips for reducing carbon emissions.
3. The system shall allow users to navigate content easily.

Module 5: User Profile and Progress

1. The system shall display user profile information.
 2. The system shall show progress based on tracked data.
 3. The system shall motivate users through achievement indicators.
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2.3 Non-Functional Requirements

1. **Usability:** The application shall be easy to use with a clear interface suitable for non-technical users.
 2. **Performance:** The application shall load screens within three seconds under normal usage conditions.
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3. Architecture and Design

ClimateBuddy follows a modular Android application architecture. The user interface is designed using XML layouts with consistent color themes representing environmental sustainability (green and neutral tones).

The app structure separates UI components, logic, and data handling to improve maintainability. Navigation between activities is designed to be simple and intuitive. Data is stored locally using structured data models.

4. System Implementation

The application is developed using **Android Studio**, with **Java** as the programming language and **XML** for user interface design. The project contains multiple Java classes for activities, data handling, and utilities, as well as XML files for layouts and resources.

Version control and collaborative development were supported through shared project folders and coordination among team members. Screenshots of UI and source code are included as evidence of implementation.

5. Testing and Validation

Testing was conducted using manual testing techniques: - Functional testing was performed to ensure each feature works as intended. - Navigation testing ensured smooth transitions between screens. - Input validation testing checked for basic error handling.

The results show that the application meets both functional and non-functional requirements.

6. Team Roles and Contributions

Each team member contributed to both development and documentation tasks. Roles included UI design, feature implementation, data handling, testing, and report preparation. Evidence of contribution is supported through code files and task distribution records.

7. References and Academic Writing

- United Nations. (2015). *Sustainable Development Goals*. <https://sdgs.un.org/>
 - Android Developers. *Android Documentation*. <https://developer.android.com/>
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Conclusion

ClimateBuddy demonstrates how mobile technology can be used to promote awareness and encourage action toward climate change mitigation. The application successfully supports SDG 13 by educating users and helping them understand their environmental impact. Future improvements may include online data synchronization and enhanced analytics.